

QUESTION-22

DIJKSTRA ALGORITHM

main.c	Output
<pre> 1 #include <stdio.h> 2 #define V 5 3 #define INF 9999 4 int main() { 5 int graph[V][V] = { 6 {0, 10, 0, 5, 0}, {0, 0, 1, 2, 0}, {0, 0, 0, 0, 4}, {0, 3, 9, 0, 2}, {7, 0, 6, 0, 0} 7 }; 8 int dist[V], visited[V] = {0}; int i, count, u, v, min; 9 for(i = 0; i < V; i++) 10 dist[i] = INF; 11 dist[0] = 0; // Source vertex 12 for(count = 0; count < V - 1; count++) { 13 min = INF; 14 for(i = 0; i < V; i++) { 15 if(!visited[i] && dist[i] < min) { 16 min = dist[i]; 17 u = i; 18 } </pre>	<pre> Vertex Distance from Source 0 0 1 8 2 9 3 5 4 7 === Code Execution Successful === </pre>

main.c	Output
<pre> 13 min = INF; 14 for(i = 0; i < V; i++) { 15 if(!visited[i] && dist[i] < min) { 16 min = dist[i]; 17 u = i; 18 } 19 } 20 visited[u] = 1; 21 for(v = 0; v < V; v++) { 22 if(!visited[v] && graph[u][v] && 23 dist[u] + graph[u][v] < dist[v]) { 24 dist[v] = dist[u] + graph[u][v]; 25 } 26 } 27 } 28 printf("Vertex\tDistance from Source\n"); 29 for(i = 0; i < V; i++) 30 printf("%d\t%d\n", i, dist[i]); 31 return 0; 32 } </pre>	<pre> Vertex Distance from Source 0 0 1 8 2 9 3 5 4 7 === Code Execution Successful === </pre>